IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to S. MITRA AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

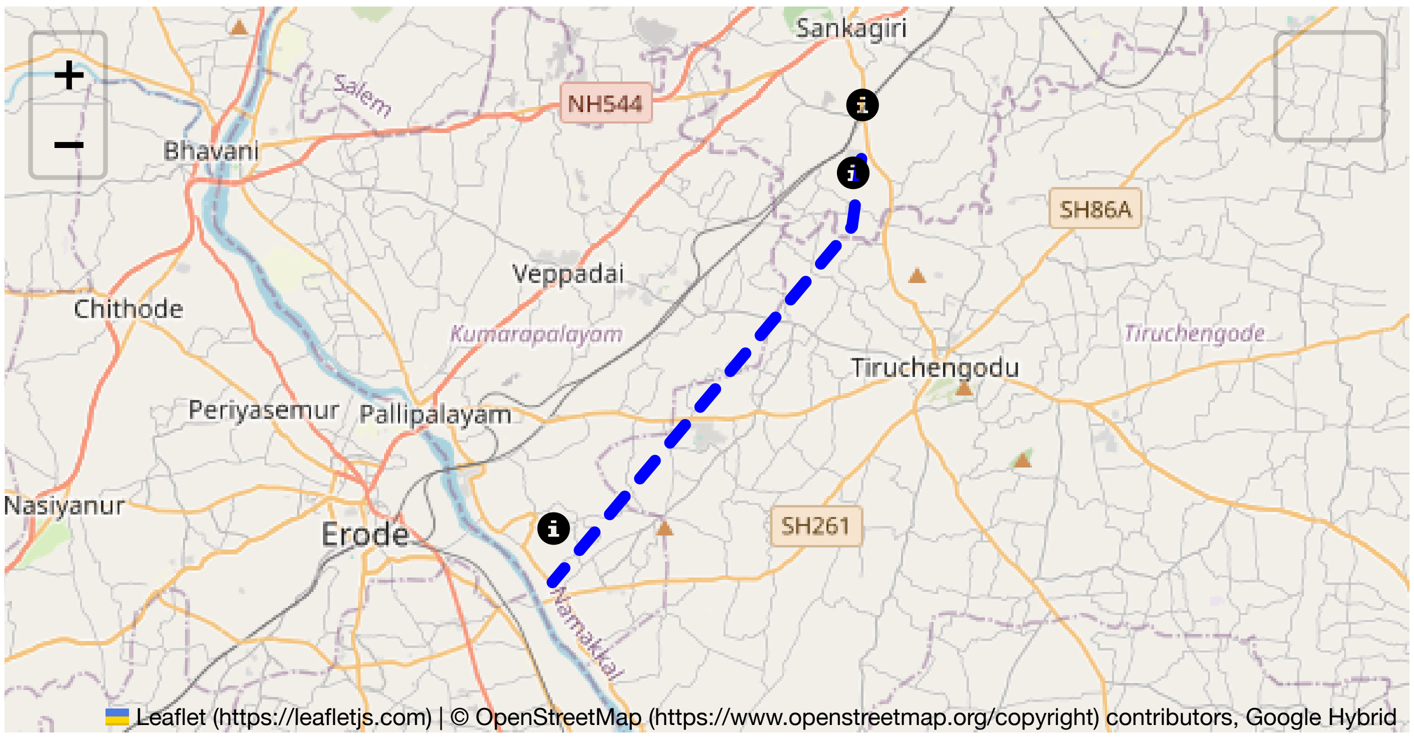
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

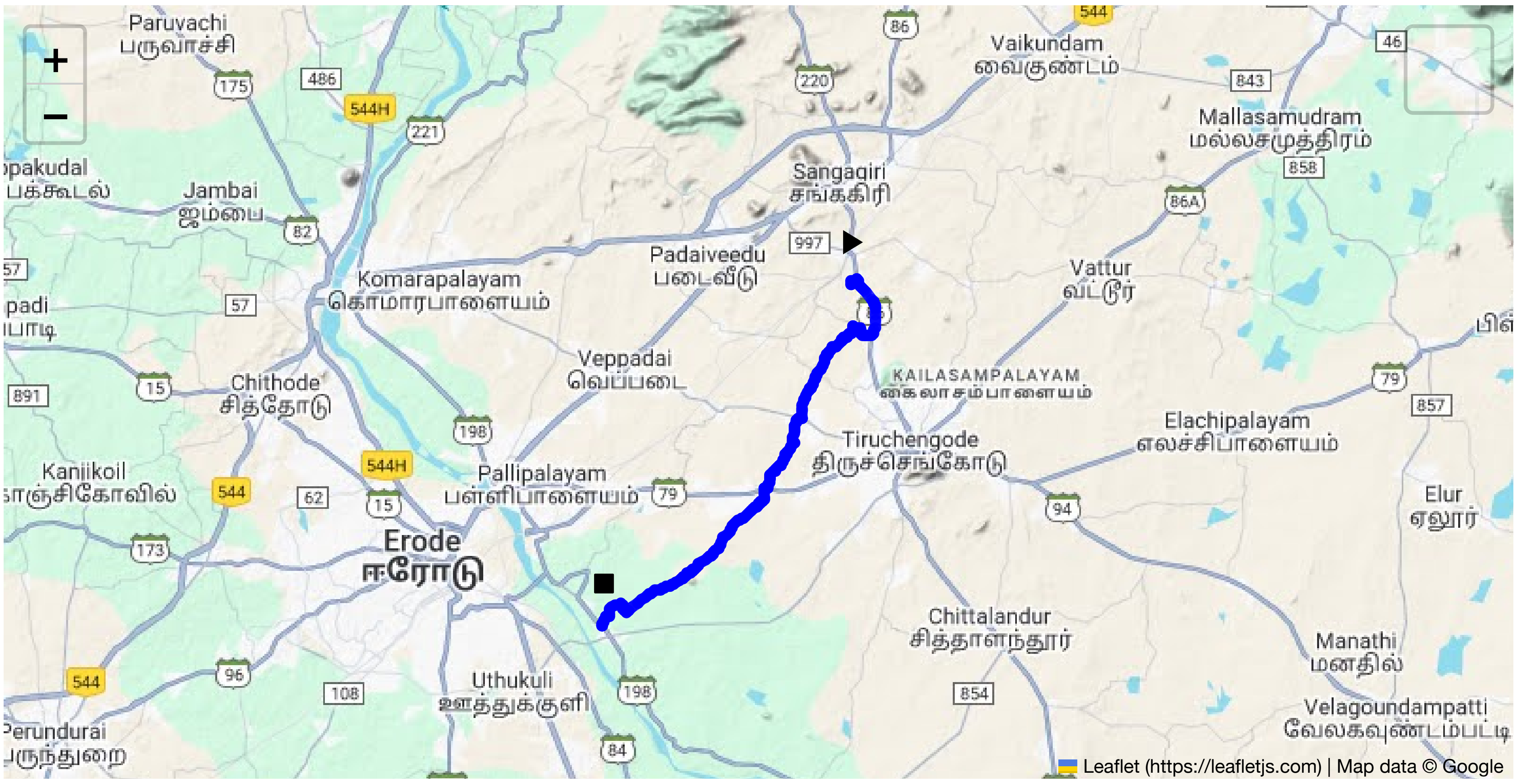
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 21.30 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.9 hours

Start: (11.4381, 77.8734)
End: (11.316777, 77.782639)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Pappampalayam via Morur spans approximately 21.30 kilometers. This journey proceeds mostly along rural highways and connecting roads typical of Tamil Nadu's regional infrastructure. The surrounding landscape includes a mix of agricultural land, small towns, and roadside commerce.

Typical Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate. The region is subject to:

- **Monsoon Season (June to September):** Heavy rains can lead to waterlogged roads and decreased visibility.
- **Summer (March to May):** High temperatures can cause road surfaces to heat up significantly, potentially leading to material degeneration or discomfort while driving.
- **Winter (October to February):** Generally mild, with minimal impact on road conditions but occasional fog in early mornings.

Traffic Patterns

- **Peak Hours:** Local peak traffic typically occurs between 8-10 AM and 5-7 PM, coinciding with commuting and school hours.
- **Congestion-Prone Areas:** The areas around Sangagiri and Morur entrances may experience delays due to local traffic density, especially near markets and schools.

Road Quality and Infrastructure

The road quality varies:

- **State Highways:** Generally well-maintained but may have narrow lanes and lack sufficient lighting at night.
- **Rural Roads:** Some sections may exhibit potholes, rough patches, or insufficient signage, necessitating cautious driving.

Alternative Routes

In case of roadblocks or emergencies:

- **Via Annur and Kanakampalayam:** While longer, this route bypasses potential congestion at town entrances.
- **Use of Local Inner Roads:** For bypassing small sections of dense traffic or roadworks, known from experience or local guidance.

Local Regulations for Hazardous Materials

- Permits are required for transporting hazardous materials.
- Strict adherence to route plans approved by local authorities.
- Restrictions may apply during festivals or specific times due to crowded festivities.

Historical Incidents

There have been minor incidents related to vehicle breakdowns and occasional collisions, particularly around busy junctions and entry points into small towns. However, significant accidents involving hazardous materials are not frequently reported.

Environmental Considerations

The route passes through agricultural zones, requiring caution to avoid contamination. Awareness of wildlife crossing is advisable, particularly in less illuminated rural sections.

Communication Coverage

- **Coverage:** Generally good in and around towns; potential dead zones in more rural stretches.
- **Telecom Services:** Reliant on major Indian network providers, varying in signal strength.

Estimated Emergency Response Times

- **Urban Areas (Sangagiri and Pappampalayam):** Estimated response within 30-45 minutes.
- **Rural Stretches (near Morur):** Response may take up to an hour due to distance and road conditions.

Overall Risk Assessment Summary

- **Risk Level:** Moderate
- **Key Risks:** Weather-induced visibility issues, road surface conditions, agricultural traffic during harvest times, and limited mobile coverage in remote areas.
- **Recommendations:** Frequent monitoring of traffic and weather updates, ensuring compliance with local transport regulations, and maintaining communication devices with satellite link support for emergencies.

By understanding these factors, drivers can better prepare for the journey, prioritizing safety and efficiency, particularly when handling hazardous materials. Regular route evaluations and updates based on local developments are advised to ensure up-to-date risk assessments.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Blind Spot	Blind Spot	11.41981, 77.88078	10 KM/Hr
5	Turn	Medium	11.42097, 77.87686	30 KM/Hr
6	Turn	Medium	11.42231, 77.87666	30 KM/Hr
7	Blind Spot	Blind Spot	11.42333, 77.87394	10 KM/Hr
8	Turn	Medium	11.42129, 77.87359	30 KM/Hr
9	Turn	Medium	11.42090, 77.87311	30 KM/Hr
10	Turn	High	11.41917, 77.87022	15 KM/Hr
11	Turn	High	11.40476, 77.86017	15 KM/Hr
12	Turn	High	11.40471, 77.86004	15 KM/Hr
13	Turn	High	11.39361, 77.85502	15 KM/Hr
14	Turn	Medium	11.39326, 77.85537	30 KM/Hr
15	Blind Spot	Blind Spot	11.39047, 77.85509	10 KM/Hr
16	Turn	High	11.39076, 77.85379	15 KM/Hr
17	Turn	Medium	11.38963, 77.85303	30 KM/Hr
18	Turn	Medium	11.38629, 77.85258	30 KM/Hr
19	Turn	Medium	11.38591, 77.85279	30 KM/Hr
20	Turn	High	11.38248, 77.85174	15 KM/Hr
21	Turn	Medium	11.38227, 77.85213	30 KM/Hr
22	Turn	Medium	11.38207, 77.85226	30 KM/Hr
23	Turn	Medium	11.37017, 77.84376	30 KM/Hr
24	Turn	High	11.36568, 77.84272	15 KM/Hr
25	Turn	High	11.36551, 77.84183	15 KM/Hr
26	Turn	Medium	11.36505, 77.84189	30 KM/Hr
27	Turn	Medium	11.36210, 77.84146	30 KM/Hr
28	Turn	Medium	11.36189, 77.84128	30 KM/Hr
29	Turn	Medium	11.36159, 77.84124	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
30	Turn	Medium	11.33668, 77.81596	30 KM/Hr
31	Turn	Medium	11.33552, 77.81470	30 KM/Hr
32	Turn	High	11.33554, 77.81355	15 KM/Hr
33	Turn	Medium	11.33489, 77.81346	30 KM/Hr
34	Turn	Medium	11.33470, 77.81336	30 KM/Hr
35	Turn	Medium	11.33160, 77.80861	30 KM/Hr
36	Turn	High	11.32143, 77.79174	15 KM/Hr
37	Turn	Medium	11.32231, 77.79039	30 KM/Hr
38	Turn	Medium	11.32344, 77.79015	30 KM/Hr
39	Turn	High	11.32376, 77.78976	15 KM/Hr
40	Blind Spot	Blind Spot	11.32358, 77.78941	10 KM/Hr
41	Blind Spot	Blind Spot	11.32423, 77.78912	10 KM/Hr
42	Turn	Medium	11.32243, 77.78555	30 KM/Hr
43	Turn	Medium	11.32013, 77.78492	30 KM/Hr
44	Turn	Medium	11.31955, 77.78388	30 KM/Hr
45	Blind Spot	Blind Spot	11.31627, 77.78274	10 KM/Hr
46	Turn	High	11.31664, 77.78249	15 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	K. S. Rangasamy College of Technology	11.3587246, 77.8278764	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.41981, 77.88078

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.42333, 77.87394

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41917, 77.87022

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.40476, 77.86017

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.40471, 77.86004

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.39361, 77.85502

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39326, 77.85537

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.39047, 77.85509

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.39076, 77.85379

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38963, 77.85303

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38629, 77.85258

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38591, 77.85279

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38248, 77.85174

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38227, 77.85213

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38207, 77.85226

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.37017, 77.84376

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.36568, 77.84272

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.36551, 77.84183

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.36505, 77.84189

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

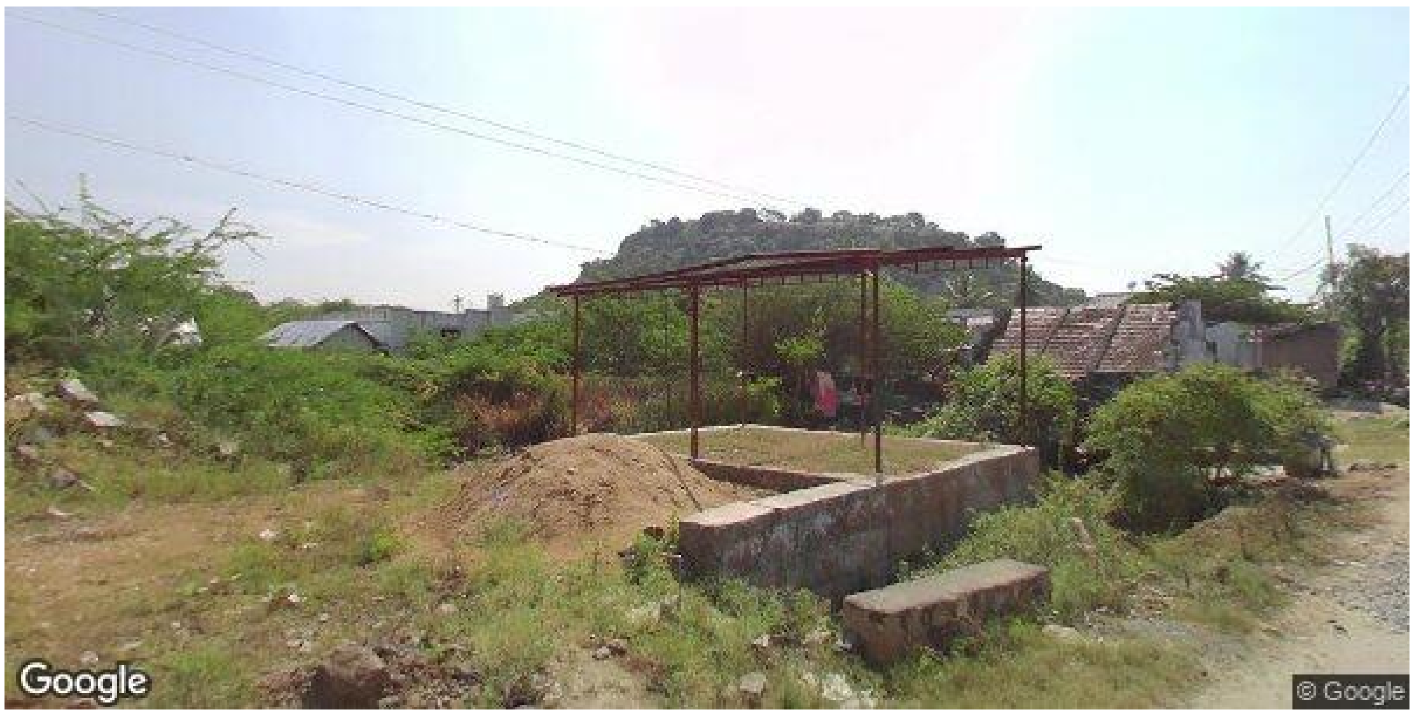
Coordinates: 11.33668, 77.81596

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33552, 77.81470



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.33554, 77.81355

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33489, 77.81346

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33470, 77.81336

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33160, 77.80861

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.32143, 77.79174

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.32231, 77.79039

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.32344, 77.79015

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.32376, 77.78976

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.32358, 77.78941

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.32423, 77.78912

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.32243, 77.78555

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.32013, 77.78492

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.31955, 77.78388

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.31627, 77.78274

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.31664, 77.78249
